

Fermi–Amaldi-Based Local Mixing Function

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I. MAIN TEXT

We propose a new local mixing function (LMF) for local hybrid functionals. This LMF is the ratio between the Fermi–Amaldi and the exact exchange energy densities:

$$\text{LMF} = \frac{e_x^{\text{FA}}}{e_x^{\text{exact}}}. \quad (1)$$

The (closed-shell) exact exchange energy density in terms of a one-electron reduced density matrix, $\gamma(\mathbf{r}; \mathbf{r}')$ is given by [1, 2]

$$e_x^{\text{exact}}(\mathbf{r}) = -\frac{1}{4} \int \frac{|\gamma(\mathbf{r}; \mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}', \quad (2)$$

where $\gamma(\mathbf{r}; \mathbf{r}') = \sum_{i=1} \varphi_i^*(\mathbf{r})\varphi_i(\mathbf{r}')$. The Fermi–Amaldi [3, 4] energy density is expressed via electron density, $\rho(\mathbf{r}) \geq 0$, and electron number, N ,

$$e_x^{\text{FA}}(\mathbf{r}) = -\frac{1}{2N} \int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}'. \quad (3)$$

The proposed LMF satisfies the *zero to one* boundary

range: both e_x^{exact} and e_x^{FA} are manifestly non-positive

$$e_x^{\text{exact}} \leq 0, \quad e_x^{\text{FA}} \leq 0, \quad (4)$$

and both are equivalent for $N \leq 2$, while for $N \geq 2$

$$e_x^{\text{exact}} \leq e_x^{\text{FA}}, \quad (5)$$

since e_x^{FA} is scaled down by N . Therefore,

$$0 \leq \frac{e_x^{\text{FA}}}{e_x^{\text{exact}}} \leq 1. \quad (6)$$

For one-electron regions, $e_x^{\text{FA}} = e_x^{\text{exact}}$, therefore, the one-electron constraint is satisfied. A particular case of a one-electron limit, i.e., the asymptotic limit ($r \rightarrow \infty$) where the exact exchange must cancel out an excessive self-repulsion of the Coulomb potential, is also satisfied:

$$\lim_{r \rightarrow \infty} \frac{e_x^{\text{FA}}}{e_x^{\text{exact}}} = 1 \quad (7)$$

since both energy densities approach $-1/2r$ as $r \rightarrow \infty$.

Another important constraint, the high-density limit,

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- [1] V. N. Staroverov, Density-functional approximations for exchange and correlation, in *A Matter of Density* (John Wiley & Sons, Ltd, 2012) Chap. 6, pp. 125–156.
 [2] R. Parr and Y. Weitao, *Density-Functional Theory of Atoms and Molecules*, International Series of Monographs on Chemistry (Oxford University Press, 1989).

- [3] E. Fermi and E. Amaldi, Le orbite ∞ s degli elementi, *Mem. Reale Accad. Italia* **6**, 117 (1934).
 [4] P. Bénilan, J. A. Goldstein, and G. R. Rieder, The fermi-amaldi correction in spin polarized thomas-fermi theory, in *Differential Equations and Mathematical Physics*, Mathematics in Science and Engineering, Vol. 186 (Elsevier, 1992) pp. 25–37.